

JOHN PHILLIP L. MALBAS

Software Developer | Ai Engineer

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PROFILE SUMMARY:

Software Developer with expertise spanning full-stack web development, data analytics, and generative AI solutions. Experienced in designing RAG pipelines with vector databases (ChromaDB), embedding LLM inference, and building custom predictive analytics platforms using Python, SQL, and Django. Skilled in transforming data into intelligent end-user products.

SKILLS:

Python, Django, CSS, Tailwind CSS, React.js, Typescript, C#, WPF, C++, Git/GitHub, JavaScript, Node.js, Html, Rest Api, MySQL, MS Test, Docker, PyTest, Ci/Cd, GitHub Action, Nginx

PROJECTS:

Retrieval Augmented Generation (RAG) Pipeline

- Engineered a domain-specific **RAG** pipeline to query the Data Privacy Act of 2012 (RA 10173), leveraging a lightweight embedding model and **ChromaDB** vector store for sub-second semantic retrieval of complex legal provisions.
- Architected a multimodal interface and stateful session layer using **Gemini** for text/voice inference and **Redis** for session memory management, ensuring persistent, low-latency conversational context across **Streamlit** web sessions.
- Streamlined legal compliance accessibility by integrating **natural language processing** and real-time document grounding, eliminating manual document navigation and accelerating privacy inquiry response times.

Gadinsight

- Designed and developed a custom data analytics platform for the BASC Gender and Development Office using **Python Django** and
- Tailwind CSS**, reducing report generation time by an estimated 50% and streamlining data tracking for over 10 office records.
- Integrated **Scikit-learn**, **Pandas**, and **Matplotlib** into the system's core backend architecture, transforming raw institutional data into actionable, visual intelligence for policy decision-making.

Custom Chess Engine

- Engineered a performant chess engine using **C#** without relying on external libraries.

Static Web Server

- Built a custom static file server from scratch using **C#** and raw TCP sockets to understand low-level networking and protocol parsing without relying on framework abstractions.
- Implemented HTTP/1.1 request handling, asynchronous stream processing, and custom security layers (IPPathSanitizer) to prevent directory traversal attacks and serve correct MIME-typed assets.
- Successfully architected a decoupled multi-threaded web server capable of rendering static web assets locally, verified through xUnit test suites and stream lifecycle testing.

Facebook Content Automation (Batang90)

- Built a config-driven prompt builder in **Python** that turns a short content idea into a structured image-generation prompt, applying JSON-defined brand rules, character references, and negative constraints to keep outputs consistent with a 90s Filipino theme.
- Integrated the **Google Gemini** image generation API with a **Streamlit** review interface, allowing approved images to be saved locally and rejected ones to be regenerated using reviewer feedback as negative prompts.
- Automated publishing of approved images to a Facebook Page through the **Graph API**, reducing manual steps between content idea, review, and posting.

PROFESSIONAL EXPERIENCE:

Full Stack Web Developer Intern

Jan 2026 – April 2026

BASC Gender and Development Office

- Architected a custom data analytics platform for the, leveraging **Django** and **Tailwind CSS** to deliver an intuitive, responsive reporting system.
- Implemented data processing and visualization features using **pandas**, **scikit-learn**, and **matplotlib** to handle demographic trends, statistical distributions, and predictive modeling profiles.
- Engineered automated data-cleaning pipelines and predictive **ML models**, enabling data-driven decision-making for over 5 years of demographic faculty records.

EDUCATION:

Bachelor Of Science In Information Technology,

2022 - 2026

Bulacan Agricultural State College

CERTIFICATIONS:

CYBER THREAT MANAGEMENT

- Studied foundational principles of **network monitoring** and **threat detection**, gaining the practical skills needed to identify system vulnerabilities and defend networks against cyber-attacks.

EXPLORING IOT WITH CISCO PACKET TRACER

- Utilized Cisco Packet Tracer to simulate and model complex IoT ecosystems, gaining the practical skills to safely connect virtual sensors, prototype automation network designs, and troubleshoot configurations.

OPERATING SYSTEMS SUPPORT

- Configured and secured major operating systems to support network infrastructure, bridging the gap between hardware and software to optimize network connectivity and system performance.